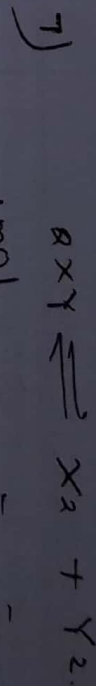
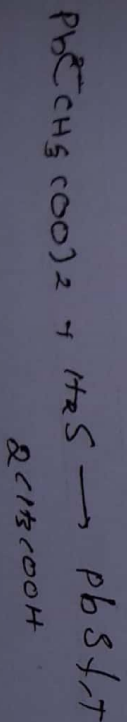
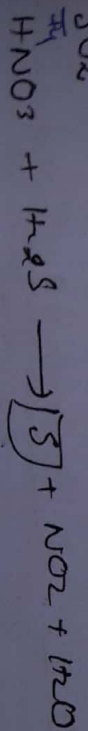


$$\text{Given HCl mol} = 0.1 \times 70 \times 10^3 \\ = 7 \times 10^3 \text{ mol}$$

$\text{NaOH} = 7 \times 10^3 \text{ mol}$

$$7 \times 10^3 - 1 \times 10^3 \\ = 6 \times 10^3 \text{ mol}$$

$$\text{Conc} = \frac{6 \times 10^3}{30 \times 10^3} = \frac{6}{30} = 0.2 \text{ mol/l}$$



$$1 \text{ mol}$$

$$0.5 \text{ mol}$$

$$0.5 \text{ mol}$$

$$K_p = \frac{P_{\text{X}_2} \cdot P_{\text{Y}_2}}{P_{\text{X}}^2 \cdot P_{\text{Y}}^2}$$

$$P_{\text{X}} =$$

$$= \frac{0.25 \times 0.25}{0.5 \times 0.5}$$

$$= \frac{1}{4} = 0.25$$

8) H-bond 2

9)

10)



$$1.52 \text{ g}$$

11) 1.52 g of C₂H₂ = 0.05 mol

$$5.94 + 2.70 = 8.64 \text{ g}$$

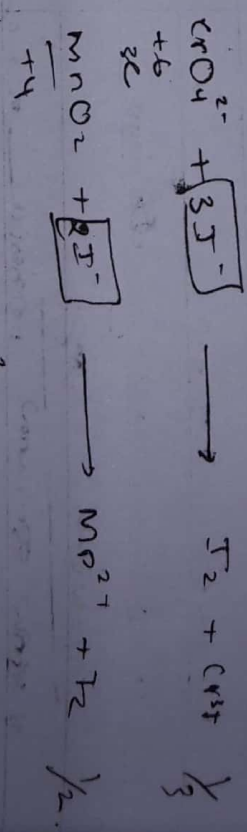
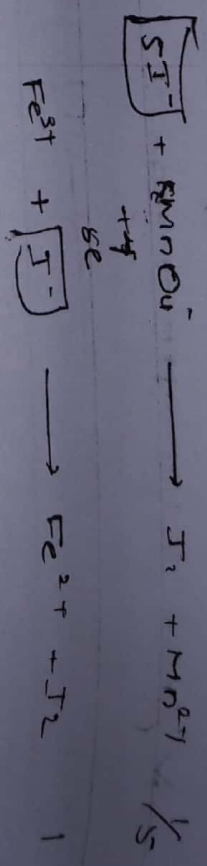
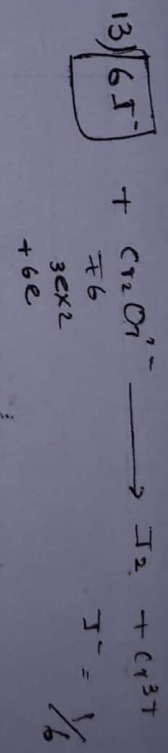
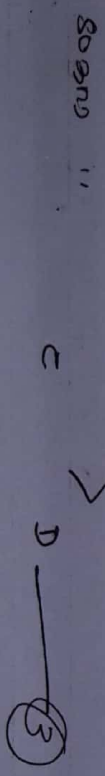
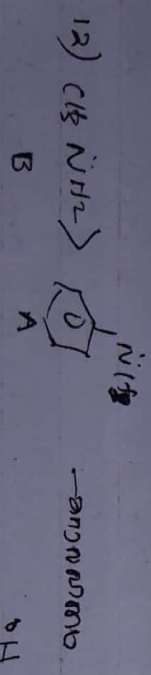
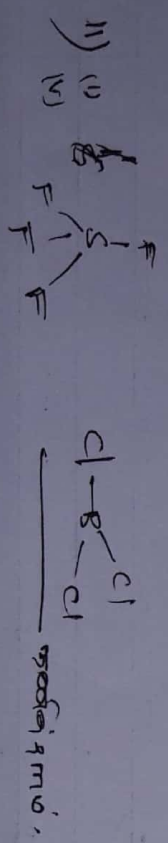
$$8.64 - 1.52 = 7.12 \text{ g}$$

(06) $0.7 \text{ CO}_2 = \frac{5.92}{44}$

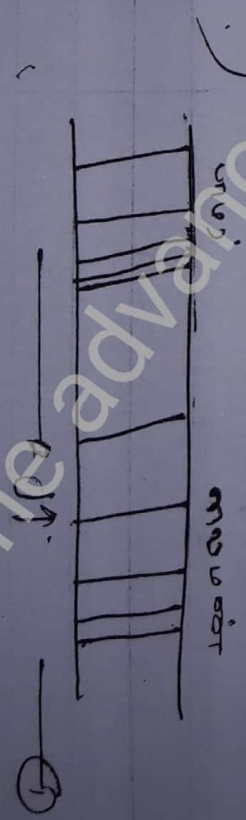
$\frac{14}{5} = \frac{2}{5.92/14.4}$

$x = 0.21 \text{ mol}$

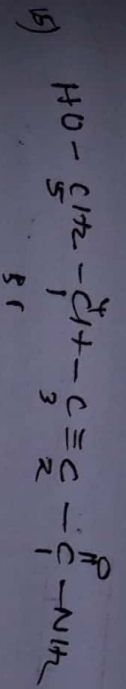
$n = m$
 $m = 0.21 \times 32$
 $= 6.72 \text{ g}$



$\frac{1}{2} = \frac{1}{2}$



As



4-bromo-5-hydroxy pent-2-ynamide
~~5-hydroxy-4-bromo-pent-2-ynamide~~

16) ശ്രേണി ക്രമം ശാസ്ത്രീയമായി തിരഞ്ഞെടുത്ത് :-

1) ദ്രവ്യഭാരം കൂടുക

2) സ്ഥിതി - യാതൊന്നും ഇല്ല

3) ദീർഘമായതും ചുരുക്കമായതും

4) കൂടുതൽ ദൃഢമായതും ദുർബ്ബലമായതും

5) ജലം കലർത്താൻ കഴിയുന്നതും കഴിയാത്തതും

6) ദ്രവ്യഭാരം കൂടുന്നതും കുറയുന്നതും

7) ദ്രവ്യഭാരം കൂടുന്നതും കുറയുന്നതും

Ex:- Na_2CO_3 , KMnO_4 ,

$\text{K}_2\text{Cr}_2\text{O}_7$, Na_2SO_4

KOH , NaOH - ദ്രവ്യഭാരം കൂടുന്നതും

കുറയുന്നതും

MgCO_3 , MgSO_4 - ദ്രവ്യഭാരം കൂടുന്നതും

17) a) Co^{2+} , H_2O കലർത്തി ലായനയ്ക്ക് കലർത്താൻ കഴിയുന്നതും

$[\text{NiCl}_4]^{2-}$, $[\text{FeCl}_4]^-$ ലെ കലർത്തൽ

b) Ca^{2+} കലർത്തി ലായനയ്ക്ക് കലർത്താൻ കഴിയുന്നതും

I^- കലർത്തി ലായനയ്ക്ക് കലർത്താൻ കഴിയുന്നതും

50% കലർത്തി ലായനയ്ക്ക് കലർത്താൻ കഴിയുന്നതും

c) $\text{Fe}(\text{OH})_3 \downarrow$ - കലർത്തൽ

18) Cu , X + H_2O $\rightarrow$ $\text{pH} \uparrow$

" + NaOH $\rightarrow$ $\text{pH} \downarrow$

H_2O Cu കലർത്തി ലായനയ്ക്ക് കലർത്താൻ കഴിയുന്നതും

NaOH " കലർത്തി ലായനയ്ക്ക് കലർത്താൻ കഴിയുന്നതും

X കലർത്തി ലായനയ്ക്ക് കലർത്താൻ കഴിയുന്നതും

H_2O , H_2 കലർത്തി ലായനയ്ക്ക് കലർത്താൻ കഴിയുന്നതും

Mg , Fe " കലർത്തി ലായനയ്ക്ക് കലർത്താൻ കഴിയുന്നതും

Fe - NaOH കലർത്തി ലായനയ്ക്ക് കലർത്താൻ കഴിയുന്നതും

Mg കലർത്തി ലായനയ്ക്ക് കലർത്താൻ കഴിയുന്നതും

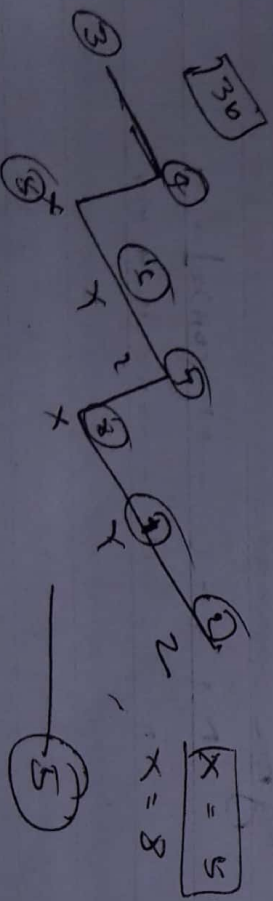
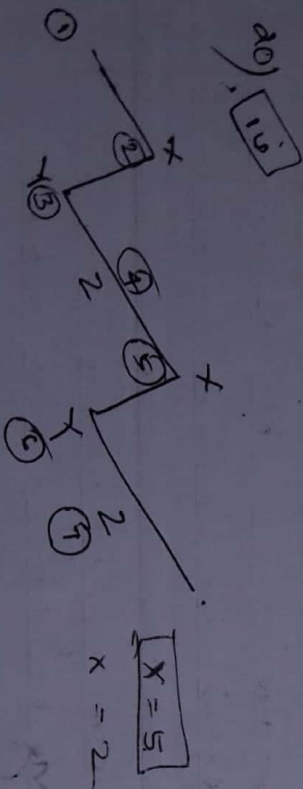
Mg + H_2O $\rightarrow$ $\text{Mg}(\text{OH})_2 \downarrow + \text{H}_2$

Mg + H_2O $\rightarrow$ $\text{Mg}(\text{OH})_2 \downarrow + \text{H}_2$

Zn + H_2O $\rightarrow$ $\text{Zn}(\text{OH})_2 \downarrow + \text{H}_2$

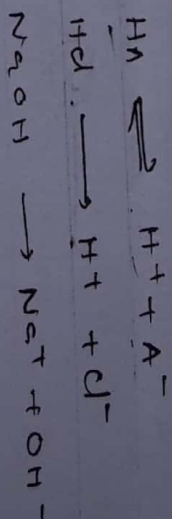
Zn - NaOH $\rightarrow$ $\text{Na}_2\text{ZnO}_2 + \text{H}_2$

but next theorem in PTh's.
but maybe 2-nd one is interesting
on graphs $\rightarrow$ PTh's.



Q11) $\frac{1}{x^2} = x^{-2}$

HA
0.02M
P 10cm²

[illegible]

2007 142

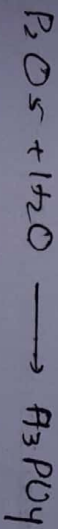
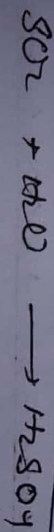
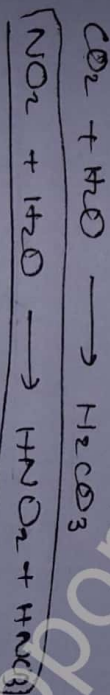
③



NO₂ ഉണ്ടായ പദാർത്ഥം ഉണ്ടാകുന്നു.
 benzene ന്റെ ഗുണമുള്ളതാണ്.
 ക്ലോറോ-ബെൻസീൻ ആണ്.
 ഓരോ ക്ലോറോ ആണ്.

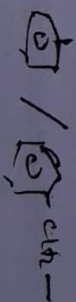
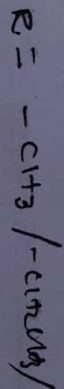
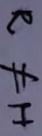
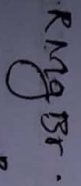
(5)

24)

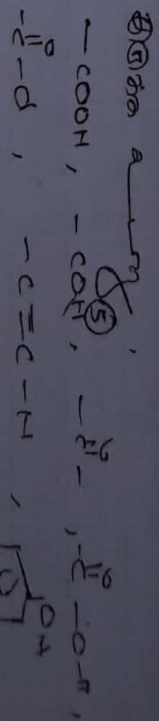


(2)

25)



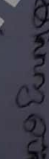
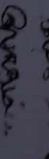
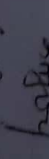
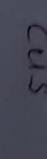
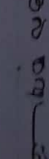
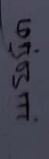
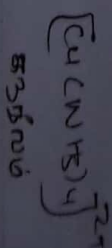
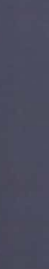
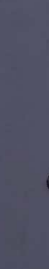
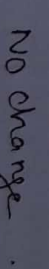
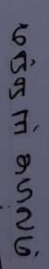
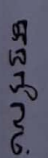
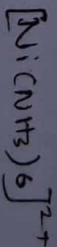
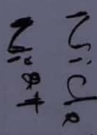
Grignard salt കേന്ദ്രം ഉണ്ടാകുന്ന ഉപയോഗം ഉണ്ടാകുന്നു.
 ഉപയോഗത്തിന് ഗുണമുള്ളതാണ്.

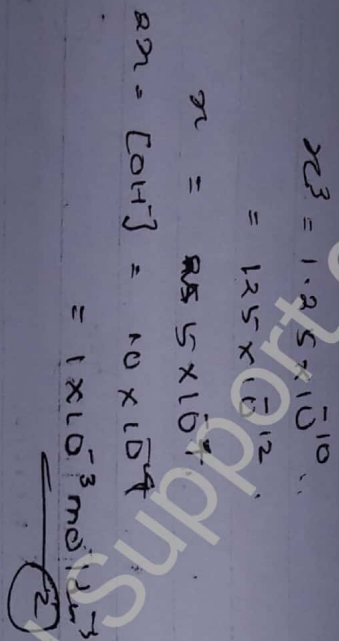
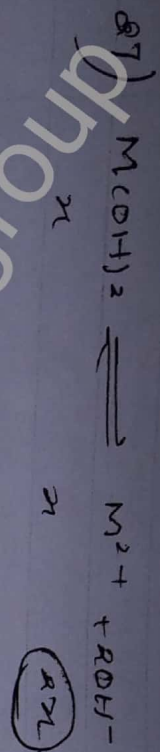


(4)

(3)

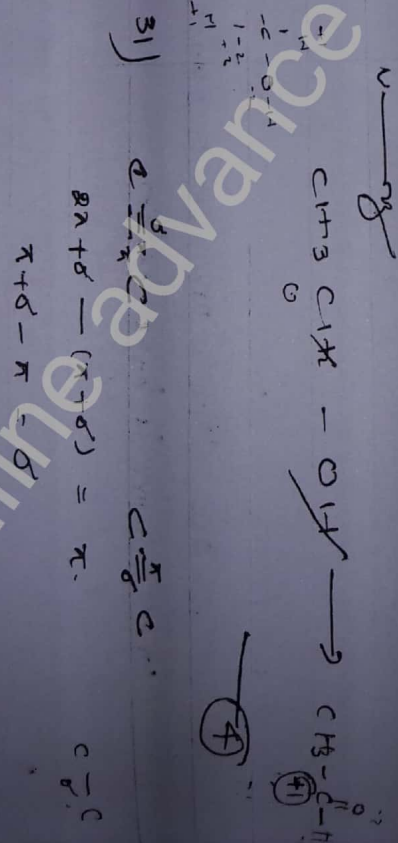
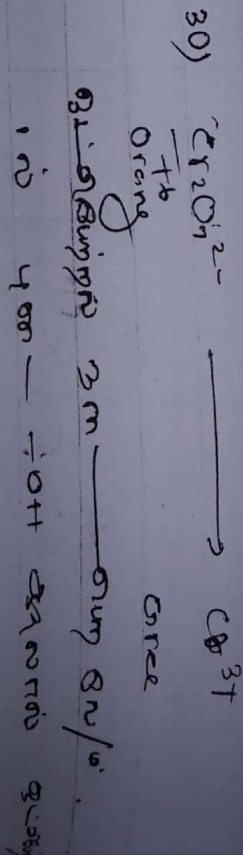
26)

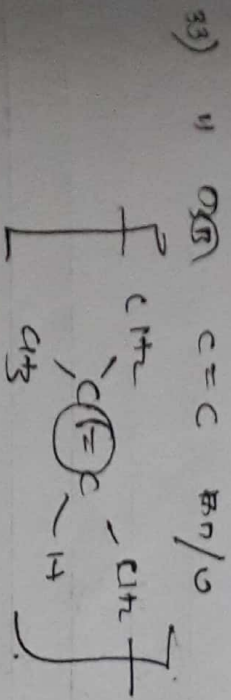




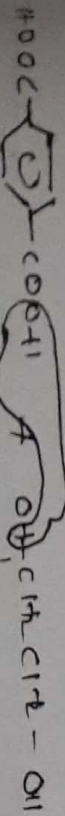
Q9) i) [T]
 2nd order reaction

ii. no
 iii. rate of reaction ↑, rate of reaction ↓
 iv. ✓ Definite rate of reaction
 v. no change





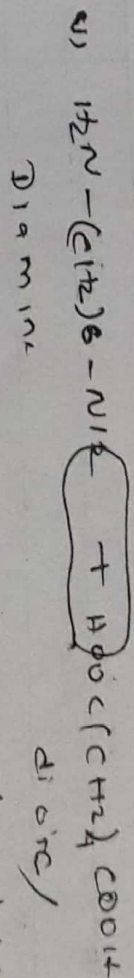
31.



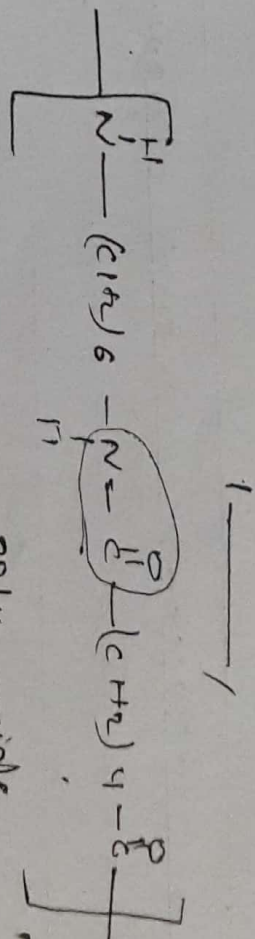
தமிழன்

~~polyester~~
polyester

n) (Amino acids) $\rightarrow$ protein.

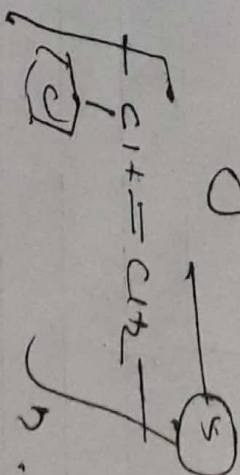


Diagonale



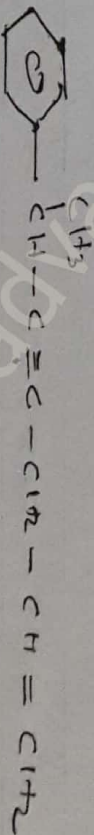
poly amide

Polystyrene



உயிர்வாழ்வு

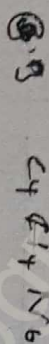
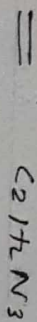
54



$\int_{\mathbb{R}^n} \delta(x) dx = 1$
 $\int_{\mathbb{R}^n} \delta(x) f(x) dx = f(0)$

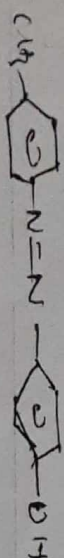
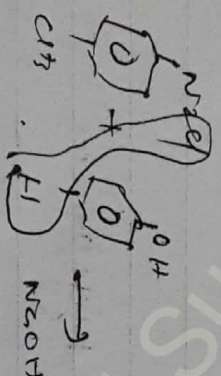
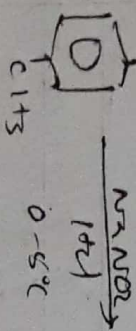
2.5

$$\begin{array}{ccccccc} \equiv & c & \lambda & c & = & c & \lambda & c & = & c & \lambda & c & - & c \\ & & & & & & & & & & & & & \\ c & & \lambda & d & & \lambda & a & & \lambda & b \end{array}$$

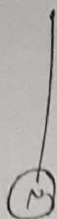
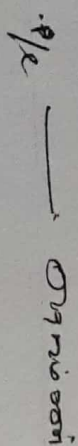
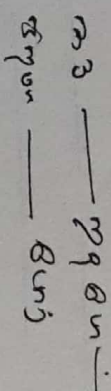
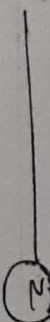



പ്രസിദ്ധ ശാസ്ത്രജ്ഞൻ

அதற்கு கர்ணியாகியும்



37) $\text{AgCl} = 1 \text{ mol} \text{ } \text{Pr}^+ / \text{C}^{\text{over}}$ $\text{Pr}^+ \text{ } \text{Pr}^{\text{mo}}$



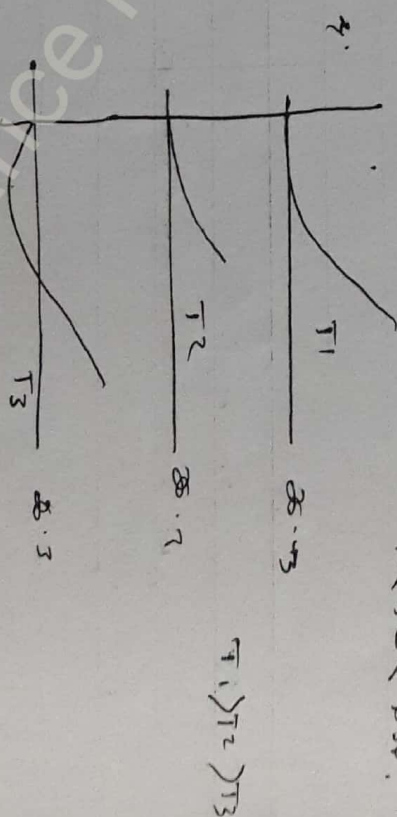
35

1. $T \uparrow$ is Burg $\Rightarrow$ Don't swarm me 3 ways

Danmings but ~~28203~~ grapha

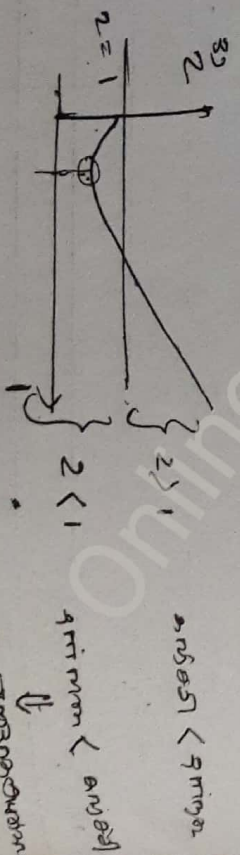
June 16

THE
GARDEN



D. The Colson 1870

உதவி
உதவி
உதவி



১৩৩৬
১৩৩৭

5

1. 2 < 1
 2. 2 > 1
 3. 2 = 1

2. அகலத்தால் எதிராக வெளவகிப்பதால்

$$P + \frac{P}{\sqrt{2}} = P$$

2017 ൽ തിരഞ്ഞെടുത്ത പട്ടികയിൽ ഉൾപ്പെടുന്നവർക്ക്
 തൊഴിലുടമയായ നിയമം അനുസരിച്ച്
 തൊഴിലുടമയായ 50 വ്യാപാരികൾക്ക്
 (V-nb).

Conditione K

amasi ON-603-97.

(w)

↑

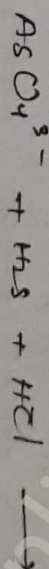
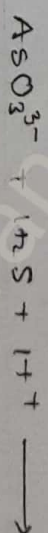
3 3

f f

[illegible]

(f) SiO_3^{2-} — orange
 AsS_3 , CdS — yellow
 $+3. \text{L}$ AsO_3^{3-}

Quay's $ASO^+ = +5$



Uniform basis upon the

43)

$$K_w = 1 \times 10^{-14}$$

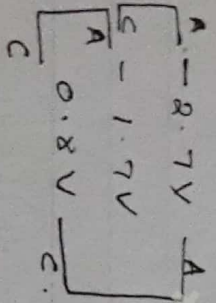
$$pH = pOH = 6$$

$$[H^+] = [OH^-] = 10^{-6} \quad \text{a) } \checkmark$$

Electrochemistry

4.

44)



$\Rightarrow$ only A anode, $\Rightarrow$ only C cathode

a) 3 different metals

b) $\checkmark$ -2.7V

c) $\checkmark$ 1.7

d) no

3

45)

a) $\checkmark$ nitric acid

b) no back titration

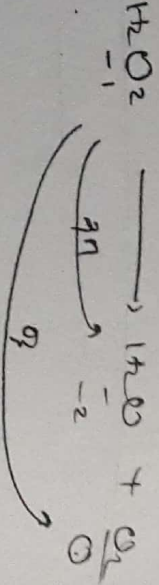
c) no

d) $\checkmark$ for

4

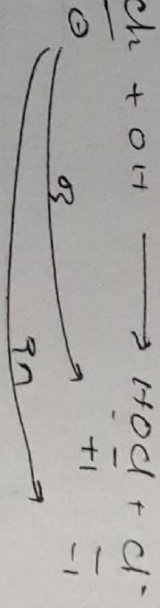
46)

a)

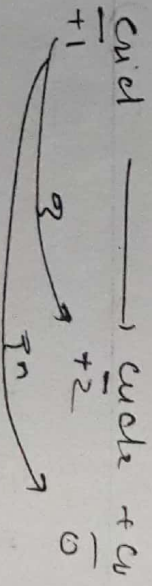


b) Oxygen gas is produced

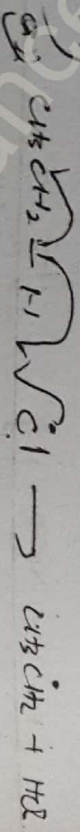
c)



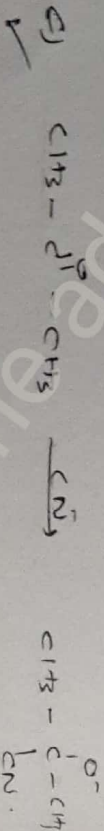
d)



47)

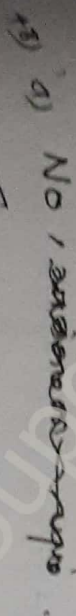


48)



49)



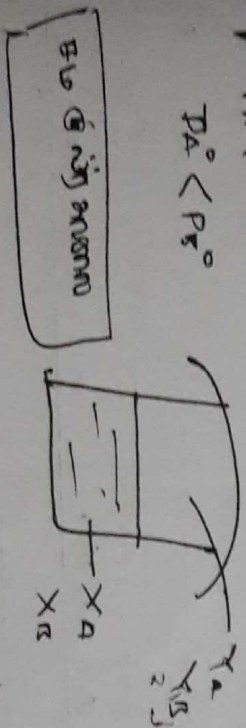


5

2005 2 17

③

49) $P_{T\alpha} > P_{T\beta}$



9. $P_A < P_B$ So the distribution is
 skewed towards the left.

$$Y_B \succ Y_A$$

PROBATION AGENCY and by B. B. B. B.

$$X \supset X_B$$

5

$$p_n = D_n^0 \alpha_n$$

$$P_{\alpha} = \frac{1}{P_{\alpha}} \times B \times$$

$$P_A = P_T \gamma_A$$

Q. 2

Qc
✓
Qc
3

PS) PA

✓ 5

10

$$P_A = P_T \cdot \gamma_A$$

$$P_A^0 n_A = (P_A^0 + P_B) y_A$$

$$P_n^0 \left(\frac{m_n}{y_n} \right) = P_n + P_S.$$

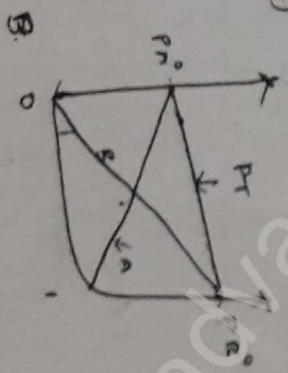
$$p_1 \leq p_2 + p_3$$

✓(c)

X B ✓
 X ✓
 X ✓
 X ✓

2/2

(or)



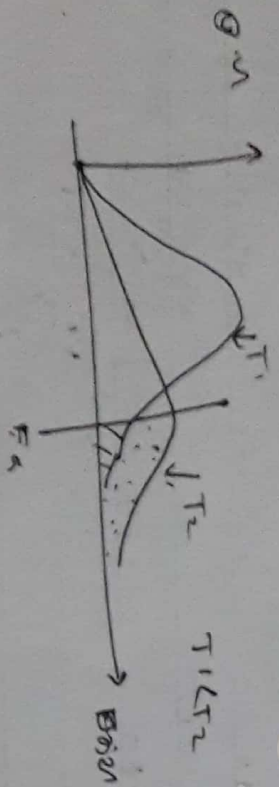
$$P_A + P_B = P_T$$

51) ബാക്സ്റ്റേർ ഏജൻസി

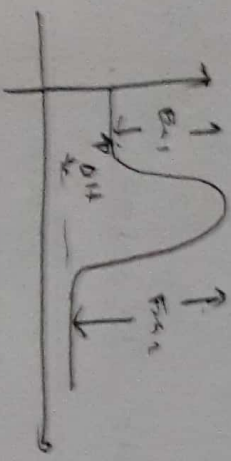
ബാക്സ്റ്റേർ $\uparrow$ - വൈദ്യുത - ഗുണ $\uparrow$
 $\downarrow$ - അപരത - $\uparrow$

✓ X
 ഇതിൽ വൈദ്യുത ഗുണ അപരത വൈദ്യുത
അപരത വൈദ്യുത

52) PbCl₂ ഉപയോഗിച്ച് വൈദ്യുത



Taas വൈദ്യുത ഗുണ അപരത



$E_H = E_{H_2} - E_{H_2}$
(H₂)

3

53) TCl₂ = 2²/8 = 2.5

Cl₂ അപരത

$NOR = 5 + 12 = 17/8 = 2.125$

X X (5)

54) pH < 7

yes cor + H₂O acid

1

55) PbCl₂ + H₂O

PbCl₂ + H₂O Pb²⁺ + 2Cl⁻

വൈദ്യുത ഗുണ അപരത

PbCl₂ + H₂O [PbCl₂]²⁺

വൈദ്യുത ഗുണ അപരത

X ✓

4

Example: benzene

normal C-H stretching energy =

11.7 eV



benzene is

normal C-H stretching energy = 2/16 = normal stretching

benzene is

T

band 8 / stretching

benzene stretching energy

$$\frac{XX}{5}$$

51)

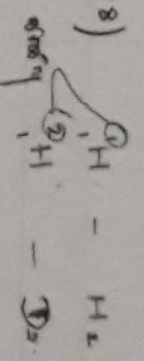
benzene oil

benzene

benzene

3

58)



n=10 p=1

$$PV = nRT$$

P ∝ n

$$P \propto \frac{m}{M} / P \propto \frac{N}{L}$$

benzene [N] [L]

benzene or / m

P same

X

$$E_k = \frac{1}{2} m v^2$$

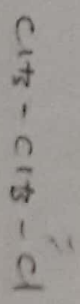
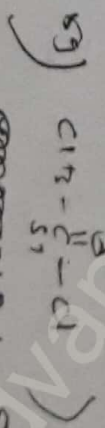
$$E_k = \frac{3}{2} kT$$

E ∝ T

$$\frac{C^2}{M} = \frac{3kT}{M}$$

benzene stretching energy

4



benzene stretching energy

benzene stretching energy

benzene stretching energy

but stretching @ 0.1 m but frequency

1